

Robin Connor Schramm

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Research Interests

My overarching goal is to enable meaningful interaction, visualization, and information dissemination, particularly through novel multimodal interfaces and technologies. Providing contextual information about users' environments interactively is a promising direction for Human-Computer Interaction (HCI), especially with the power of contextual AI. My multidisciplinary research in HCI is deeply rooted in experimental methods. A significant portion of my work involves conducting in-car field studies to evaluate innovative interaction paradigms and visualizations using empirical evidence. These studies are crucial as they offer real-world insights and data, ensuring that the developed systems are effective and practical in actual conditions beyond the lab. My current work primarily involves designing, implementing, and testing novel interactive systems that have the potential to shape future mobility.

Education

University College London (UCL)

Mar. 2026 – May 2026

Visiting Researcher

Host: [Dr. Mark Colley](#)

Support: by the German Academic Exchange Service (DAAD) [HAW.International](#) program.

RheinMain University & Mercedes-Benz Tech Innovation

Dec. 2022 – present

PhD in Human Computer Interaction

Advisor: [Prof. Dr. Ulrich Schwanecke](#)

Thesis Title: Interactive Points of Interest for In-Car Augmented Reality

Reutlingen University

Sep. 2019 – Sep. 2021

Master of Science in Human-Centered Computing

Thesis Title: Simulation of Long-Term SLAM for Mobile Robots

Thesis Cooperation: [Fraunhofer IPA](#)

Overall Grade: 1.3 - A equivalent

Reutlingen University

Mar. 2016 – Aug. 2019

Bachelor of Science in Computer Science and Media

Thesis Title: Comparison of Motion Tracking and Inverse Kinematics for Presence in Virtual Reality

Thesis Cooperation: [Fraunhofer IAO](#)

Job Experience

Software Engineer

Oct. 2021 – present

Mercedes-Benz Tech Innovation

Böblingen, Germany

I develop prototypes in fast-changing environments with evolving hardware requirements. Specializing in creating in-car Augmented Reality applications using Unity, I focus on developing novel interactive systems that enhance user experience. My work also includes thorough testing in labs, in the field, and through empirical studies. Additionally, I create showcases and demos to demonstrate the capabilities of these systems. Collaborating with a team to drive innovation and immersive technologies across the company, I deliver efficient and client-focused solutions, ensuring smooth integration and optimal functionality within advanced mobility systems.

Research Assistant

Feb. 2021 – Sep. 2021

Fraunhofer IPA

Stuttgart, Germany

I developed a virtual twin-like simulation for visual SLAM for mobile robots using Unity. This involved creating realistic simulations to test and refine visual SLAM algorithms, ensuring they perform effectively in dynamic and changing environments.

Research Assistant

Reutlingen Research Institute

Mar. 2020 – Feb. 2021

Reutlingen, Germany

In the [Project KI Delta Learning](#), I developed Unity simulations using photogrammetry data and applied computer vision techniques to generate synthetic data for autonomous driving.

Research Assistant

Fraunhofer IAO

Aug. 2019 – Feb. 2020

Stuttgart, Germany

I developed innovative tools for collaborative Virtual Reality applications, specifically tailored for construction data visualization and analysis.

Working Student

Bizerba

Feb. 2018 – Jan. 2019

Balingen, Germany

I developed user interfaces for embedded systems and created new Linux build pipelines for long-term support systems.

Teaching

Lecturer - Scientific Writing

Reutlingen University

Oct. 2023 – present

Reutlingen, Germany

As a Lecturer for the seminar "Topics in Computer Science" I guide students through the basics of the scientific process, citation practices, systematic literature reviews, and academic writing. I provide detailed feedback on their papers and teach the essentials of delivering research talks and using AI tools. I have reviewed and graded papers and talks of over 150 students over the course of five semesters.

Tutor - Basics in Programming

Reutlingen University

Mar. 2019 – Feb. 2021

Reutlingen, Germany

Led the Programming Fundamentals Lab, where I taught students the basics of programming in C and Python through interactive live coding sessions, comprehensive reviews, and engaging Q&A discussions. Additionally, I reviewed exam questions and assisted in grading coding exams.

Publications

1. **Robin Connor Schramm**, Ginevra Fedrizzi, Markus Sasalovici, Jann Philipp Freiwald, and Ulrich Schwanecke. 2025. *Augmented Journeys Interactive Points of Interest for In-Car Augmented Reality*. In CHI Conference on Human Factors in Computing Systems (CHI '25), ACM, doi: [10.1145/3706598.3714323](https://doi.org/10.1145/3706598.3714323)
🏆 Received a **CHI Honorable Mention Award for Best Paper** (top 5%)
2. **Robin Connor Schramm**, Markus Sasalovici, Jann Philipp Freiwald, Michael Otto, Melissa Reinelt, and Ulrich Schwanecke. 2025. *Blending the Worlds World-Fixed Visual Appearances in Automotive Augmented Reality*. In CHI Conference on Human Factors in Computing Systems (CHI '25), ACM, doi: [10.1145/3706598.3713185](https://doi.org/10.1145/3706598.3713185)
3. Markus Sasalovici, Albin Zeqiri, **Robin Connor Schramm**, Oscar Javier Ariza Nunez, Pascal Jansen, Jann Philipp Freiwald, Mark Colley, Christian Winkler, and Enrico Rukzio. 2025. *Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles*. In CHI Conference on Human Factors in Computing Systems (CHI '25), ACM, doi: [10.1145/3706598.3714077](https://doi.org/10.1145/3706598.3714077)
4. **Robin Connor Schramm**, Markus Sasalovici, Axel Hildebrand, and Ulrich Schwanecke. 2023. *Assessing Augmented Reality Selection Techniques for Passengers in Moving Vehicles A Real-World User Study*. In ACM International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '23), ACM, doi: [10.1145/3580585.3607152](https://doi.org/10.1145/3580585.3607152)
5. Markus Sasalovici, Stephan Leenders, **Robin Connor Schramm**, Jann Philipp Freiwald, Hannes Frederic Botzet, Daniel Keßelheim, Thomas Krach, and Christian Winkler. 2023. *In-Car Office: Can HMD-Based AR Alleviate Passenger Motion Sickness?*. In 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI '23 Adjunct), ACM, doi: [10.1145/3581961.3609869](https://doi.org/10.1145/3581961.3609869)

Service and Volunteering

- Associate Chair for the [NordiCHI'26](#) Late-Breaking Work track
- Associate Chair for the [MuC'26](#) Full Papers track
- Part of the Organizing Committee as Proceedings Chair for [MUM'26](#)
- I regularly review papers at top HCI venues such as CHI, MobileHCI, AutomotiveUI, ACM ISS, and the International Journal of Human-Computer Studies
- Student Volunteer at IEEEVR '21

Thesis Supervision

Master Theses as main supervisor (all at Mercedes-Benz Tech Innovation):

- Leo Kruse, University of Porto (2025)
- Alireza Parchami, Saarland University (2025)
- Hongcheng Jia, University of Stuttgart (2024)
- Ginevra Fedrizzi, University of Trento (2024)

Talks, Seminars, and Demos

- Ulm University (2025): "In-Car Mixed Reality - current research and challenges" - invited talk at the Automotive User Interfaces Lecture
- Mercedes-Benz Tech Innovation (2025): "Success Story - Research in Automotive User Interfaces" - in-person presentation at the company townhall meeting
- Mercedes-Benz PhDs poster session [Doktoranden Marktplatz](#) (2024) - in-person
- WeAreDevelopers World Congress - [Mercedes-Benz booth](#) (2023): "In-Car Augmented Reality with the Varjo XR-3" - in-person interactive live demonstration
- Reutlingen University (2023): "Automotive User Interfaces in HCI" - in-person
- Mercedes-Benz AG (2023): "Augmented Reality Selection Techniques in Moving Vehicles" - digital research cluster talk
- Mercedes-Benz Tech Motion (2022): "Integration of Head-Mounted Displays (HMDs) in Cars" - in-person tech talk

Skills

- **Coding:** Mainly *C#* for prototyping and creating interactive XR applications and user-studies, especially with Unity for both Windows and Android; *R* for statistical analysis and data visualization; also proficient in *C*, *Python*, and *Java*
- **AI-Assisted Development:** Proficient in leveraging *GitHub Copilot* and *Codex* for code generation, refactoring, debugging, and technical documentation, with rigorous validation of outputs for quality and reliability
- **Industry Projects:** Proficient in working in teams with agile frameworks like Scrum and managing tasks with Kanban boards and Jira; regularly use and maintain *GitHub* repositories
- **Research:** Planning and conducting *user studies*, especially in the field; *Systematic Literature Review* for comprehensive analysis and synthesis of existing research; analyzing and interpreting both *qualitative* and *quantitative data*; *Writing and Reviewing* research papers in HCI and Computer Science
- **Languages:** *German* – Native; *English* – Proficient